COSTS ASSOCIATED WITH AGRICULTURAL STRUCTURES

Introduction

Cost associated with agricultural structures can be put under five main groups:

- Initial cost
- Maintenance and repair cost
- Insurance
- Taxes
- Interest on investment

Initial cost of structure

The initial cost includes all cost associated with the construction process of the structure. The main components of the initial cost are:

The construction/constructional cost which includes:

- material cost
- labour cost
- transportation cost
- consultation or consultancy cost
- contingency cost

Initial building cost may be estimated in a total cedi quantity or be expressed in other ways eg. the following ways.

- 1. Cost per animal unit

The cost per animal unit multiplied by the number of animals gives the initial building cost.

- 2. Cost per unit volume

No general cost data are provided to estimate the initial building cost, since this cost varies so widely from place to place.

Material cost

Material cost is the major cost in the construction of a structure and is determined by:

- volume
- area
- linear feet
- pieces
- bags, etc

depending on the standard unit of measure for the materials involved

Example

- Cement is measured in bags
- Wood is measured in feet
- Roofing sheets measured by area

To be able to assess the material cost, there would be the need to list all the materials involved and prepare a bill of quantities.

What is a bill of quantities?

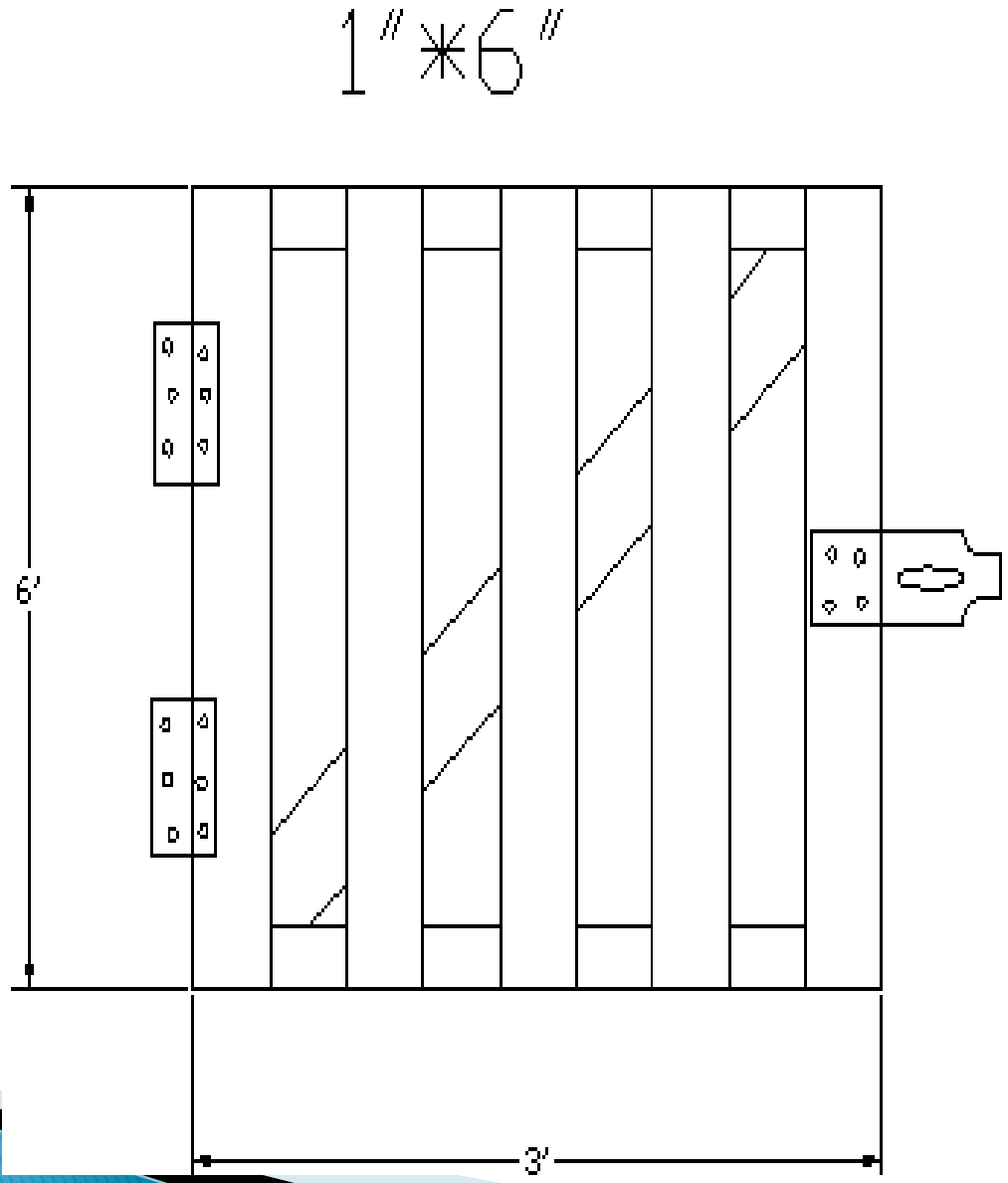
A bill of quantities

A bill of quantities is a list of the amount of all the materials required for the construction of a structure and the cost involved arranged in a logical order.

The following terms are involved in the preparation of a bill of quantities:

- Item/part name
- Number of pieces
- Type of material
- Size of pieces
- Description of materials
- Total amount/quantity of material
- Unit cost
- Total cost

Example: Consider the sketch of a farm gate given below and determine the material cost:



- The following information is given

Unit cost of Wawa timber 1” x 12” x 12”	¢20.00
Pair of hinges (4”)	¢3.00
Pair of Hasp (4”)	¢5.00
Screw (1½”) - per piece	¢0.50
Screw (2”) - per piece	¢0.70
Nails (½”) - per pound	¢2.00
Nails (2”) - per pound	¢2.00
Paint (Gloss) -1litre	¢10.00

Bill of quantities

Item	No. Of pieces	Type of material	Size	Description of material	Quantity of material	Unit cost (¢)	Total cost (¢)
Board	8	Wood	1”x6”x6’	Wawa	2	10.00	20.00
Hinges	1 pair	Steel	4”	mild steel	pair	3.00	3.00
Hasp	1 pair	Steel	8”	mild steel	pair	5.00	5.00
Nails	½ pound	Steel	1 ½”	mild steel	1Ib	2.00	2.00
Nails	½ pound	Steel	2”	mild steel	1Ib	2.00	2.00
Screw	15	Steel	1 ½”	mild steel	15	0.50	7.50
	10	Steel	2”	mild steel	10	0.70	7.00
Paint	-	Gloss	-	Brown	2 litres	10.00	20.00

The material cost is the sum of all the subtotal cost in the last column.

Note:

For Buildings, the bill of quantities can be done according to:

- **The main operations:** site clearing, excavation and filling and the external works (walls etc)
- **Type of trade:** (carpentry, masonry work, painting, electrical or plumbing work).

Labour

- Labour assessment depends on the community (e.g. town or village) and the type of project.

It may be considered as:

- a percentage of the total material cost
- by contract or
- on per day basis

Maintenance and repair cost

- Buildings deteriorate due to age, weathering and use.
- This makes maintenance and repair necessary at all times to maintain the building's appearance and serviceable condition.
- The cost of periodic maintenance and repair of a building is usually considered on annual basis.

- The amount set aside for maintenance and repair is always a percentage of the initial cost.
- This ranges between 1% and 3% of the initial cost.
- Here the mode of construction is an important factor that is considered.

The following maintenance measures are necessary:

- The use of suitable materials for the prevailing climatic conditions
- Inspection of buildings and structures regularly
- Prompt action on maintenance
- Use of parts that are easily replaceable
- Cleaning, repainting, reproofing, replacing, and or repair of broken parts

Taxes

- This is usually taken as the property tax that must be paid on a building.

- Property tax varies from community to community. However a reasonable estimate would be 1 – 2% of the initial building cost every year.

Insurance

- Because of risk from fire, floods, accidents etc., insurance is necessary.

- An estimate of annual insurance cost could be about 1 – 2% of cost of initial building cost.

Interest on investment

- Interest is included as a building or structure cost since money that is tied up in a building or structure is not available for investment or use elsewhere.
- The prevailing interest rate for borrowing or lending is the value or figure to use.

Estimate of initial building cost and total annual building cost

First determine the following information and their amounts:

- Taxes (T)
- Insurance (I)
- Repair and maintenance (R&M)
- Interest on investment (i)

Annual cost method

- $$CRF = \frac{i(1+i)^n}{(1+i)^n - 1}$$

- CRF = (capital recovery factor)

where i is the interest rate

n is the recovery period

Procedure used to estimate annual building cost

STEPS:

- Determine or estimate the initial building cost.
- Using an appropriate value for interest rate and assuming a value for the expected building life (years of recovery) estimate the value for the annual principal and interest payment percentage or factor.

- Sum up the CRF factor or percentage value and the annual % value for taxes, insurance, and maintenance and repairs to obtain the overall annual cost percentage

  %Cost = CRF + T + I + M&R

- The total or overall annual cost % multiplied by the initial building cost gives a total annual building cost.

Example:

- Calculate the total annual building cost for an initial building cost of ¢9.6m for a building life of 20 years and 8% interest rate.Take taxes 1%, insurance 1% and 2% for repair and maintenance of the initial building cost.

Solution

Initial building cost = ¢9,600,000

Interest rate (i) = 8%

Interest period (n) = 20 yrs

Interest rate (i) and Interest period (n) are used to determine the capital recovery factor (CRF)

$$CRF = \frac{i(1+i)^n}{(1+i)^n - 1}$$

$$= \frac{0.08(1+0.08)^{20}}{(1+0.08)^{20} - 1} = 10.2\%$$

- Total annual cost % will be the sum of all the % components for
  - Interest on investment (CRF)
  - Taxes
  - Insurance
  - Repair and maintenance

Total annual cost % = (10.2 + 1 + 1 + 2) %

= 14.2 %

Annual cost = 0.142 x ¢9.6m

= 1.36m

Economic feasibility

When the total annual cost of a building or structure is compared to the expected income, it forms the basis for deciding whether or not the building or structure is worth constructing.